

VastraCo

Secure E-Commerce Platform

Using AWS KMS, Secrets Manager & Amazon S3

AWS KMS

Secrets Manager

Amazon S3

IAM Roles

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Introduction

VastraCo is a microservices-based e-commerce application developed to provide secure and scalable online shopping functionality. The application consists of User Service, Product Service, and Order Service, each backed by its own PostgreSQL database. To improve security and cloud-native architecture, AWS services such as AWS Key Management Service (KMS), AWS Secrets Manager, and Amazon S3 were integrated into the application.

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Business Requirement

The application requires secure management of sensitive information such as JWT secrets and database passwords. Additionally, product images must be stored in a scalable and highly available storage solution instead of relying on external image providers. To address these requirements:

- **AWS KMS** is used to encrypt sensitive information.
- **AWS Secrets Manager** is used to securely store application secrets.
- **Amazon S3** is used to store and serve product images.
- **IAM Role-based authentication** is used to securely access AWS services without hardcoded access keys.

3 Architecture Diagram

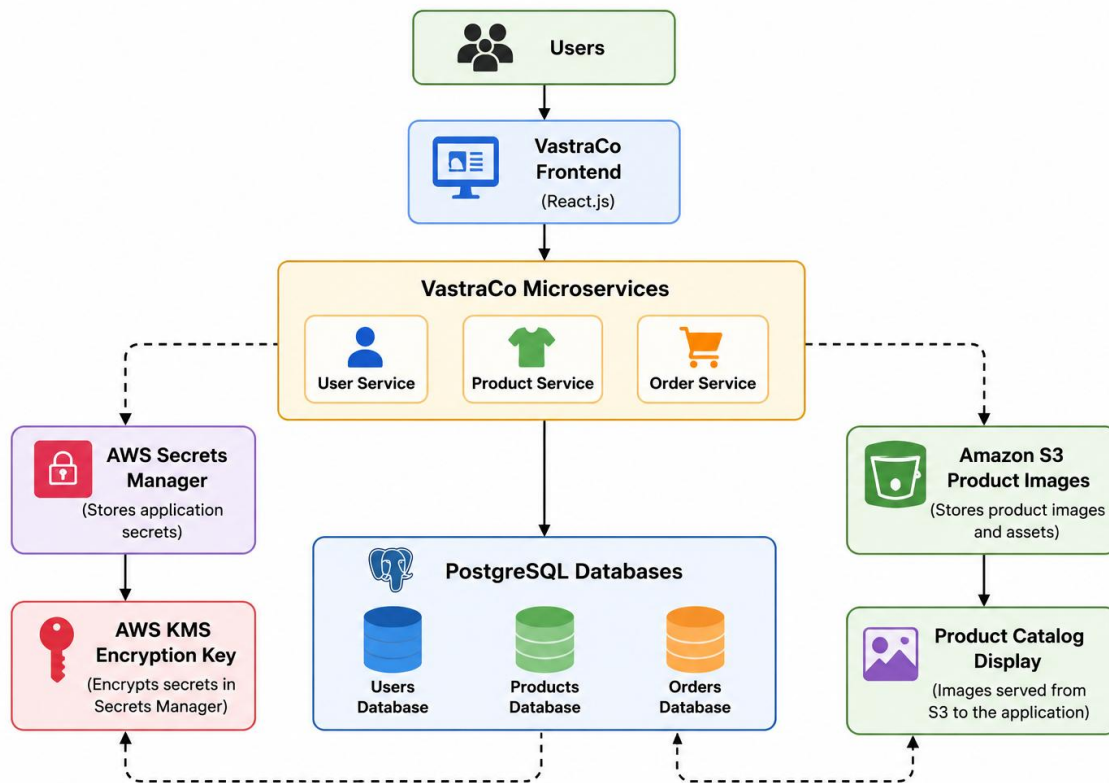


Figure 1 – VastraCo AWS Architecture: Microservices, Secrets Manager (KMS-encrypted), S3, IAM Role Auth, PostgreSQL Databases

4 AWS KMS Implementation

AWS Key Management Service (KMS) was used to create a customer-managed encryption key for securing application secrets.

Observation: The KMS key was successfully created and enabled. The key is actively used for encrypting application secrets stored in AWS Secrets Manager.

General configuration

Alias vastraco-kms-key	Status Enabled
ARN  arn:aws:kms:us-east-1:368740523957:key/e2b3d2ab-0763-449b-88de-be9e21361780	Description KMS key for VastraCo application secrets
Current key material ID bab6116a1a525f148b99a04ded73775707a344598fcbfc64082d123ce98f6879	
Creation date Jun 05, 2026 17:34 GMT+5:30	
Last used 1 hour ago	
Regionality Single region	

5 AWS Secrets Manager Implementation

AWS Secrets Manager was used to securely store application credentials and sensitive configuration data.

☰ [AWS Secrets Manager](#) > [Secrets](#) > [vastraco/application](#) ⓘ ⚙

vastraco/application

Secret details

Encryption key
 vastraco-kms-key

Secret name
 vastraco/application

Secret ARN
 arn:aws:secretsmanager:us-east-1:368740523957:secret:vastraco/application-o08XNh

 **Actions** ▼

Secret description
 Application secrets for VastraCo microservices

Secret type
-

Secret value Info
Close

Retrieve and view the secret value.

Key/value	Plaintext
Secret key	Secret value
jwt_secret	vastraco_super_secret_jwt_key_change_in_prod
user_db_password	users_pass_123
product_db_password	products_pass_123
order_db_password	orders_pass_123

Observation: All application credentials are stored securely and encrypted using the customer-managed KMS key.

6 Secret Retrieval Validation

To verify successful integration, the EC2 instance retrieved the secret using IAM Role-based authentication.

Command Used

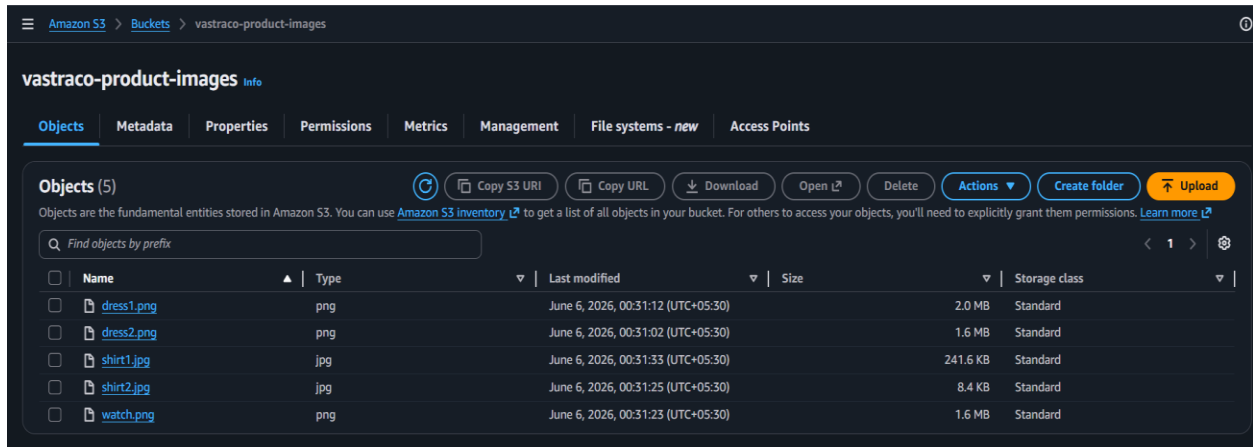
```
root@ip-172-31-40-192:/home/ubuntu# aws secretsmanager get-secret-value \
--secret-id vastraco/application \
--region us-east-1
{
  "ARN": "arn:aws:secretsmanager:us-east-1:368740523957:secret:vastraco/application-o08xNh",
  "Name": "vastraco/application",
  "VersionId": "fa7ad756-af5c-4b9c-b288-ed89628c5cf0",
  "SecretString": "{\r\n  \"jwt_secret\": \"vastraco_super_secret_jwt_key_change_in_prod\",\r\n  \"user_db_password\": \"users_pass_123\",\r\n  \"product_db_password\": \"products_pass_123\",\r\n  \"order_db_password\": \"orders_pass_123\"\r\n}",
  "VersionStages": [
    "AWSCURRENT"
  ],
  "CreateDate": "2026-06-05T12:22:55.323000+00:00"
```

Observation: The application successfully retrieved encrypted secrets from AWS Secrets Manager without using AWS Access Keys or Secret Keys.

7 Amazon S3 Integration

Amazon S3 was integrated to store and manage product images for the VastraCo application.

S3 Bucket Configuration

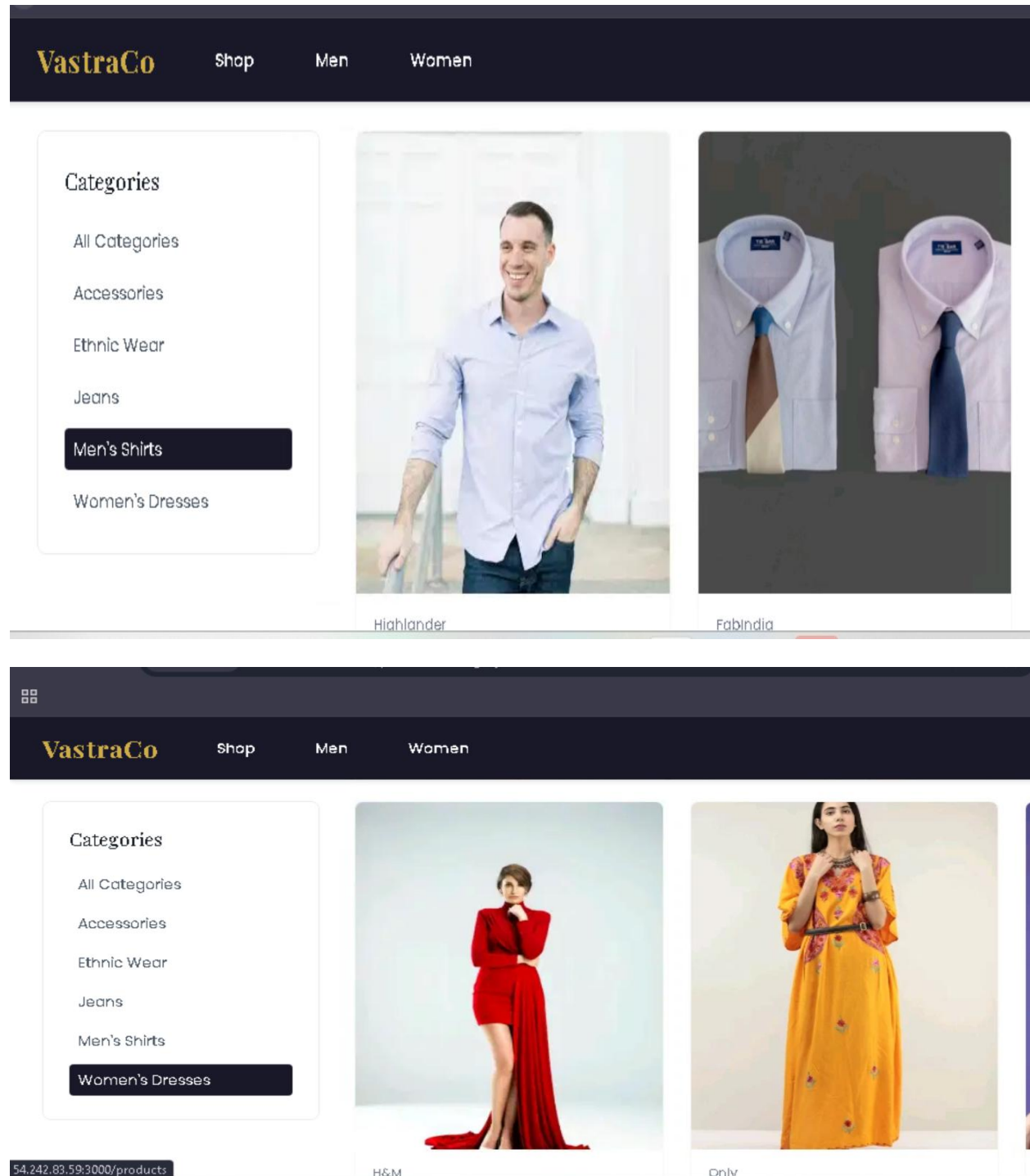


Benefits

- Highly scalable object storage
- High durability (99.999999999%) and availability
- Centralized image management
- Cost-effective storage for static assets

8 Product Image Hosting Through S3

Product images previously referenced from external image sources were migrated to Amazon S3. The application now loads product images directly from the S3 bucket, ensuring consistent performance and availability.



Observation: The application successfully retrieves and displays product images hosted in Amazon S3, replacing all external image dependencies.

9 Security Benefits

Service	Benefit	Detail
AWS KMS	Centralized Encryption	Customer-managed keys with full audit trail and rotation support
AWS KMS	Key Isolation	Dedicated vastraco-kms-key ensures separation from other workloads
Secrets Manager	No Hardcoded Credentials	All DB passwords and JWT secrets stored outside application code
Secrets Manager	Encrypted at Rest	Secrets encrypted using KMS customer-managed key
IAM Roles	No Access Keys on Servers	EC2 instances use temporary credentials via instance profiles
IAM Roles	Least-Privilege Access	Each service has only the minimum permissions required
Amazon S3	Durable Asset Storage	11-nines durability for product images with versioning support
Amazon S3	Simplified Management	Single source of truth for all product media assets

10 Conclusion

The VastraCo application was successfully enhanced using AWS cloud security and storage services. The following key outcomes were achieved:

- **AWS KMS** – Centralized encryption with a dedicated customer-managed key (vastraco-kms-key) protects all sensitive data at rest.
- **AWS Secrets Manager** – All credentials (JWT secret, three DB passwords) are stored encrypted, eliminating hardcoded secrets from application code.
- **Amazon S3** – Product images migrated from external providers to vastraco-product-images bucket, improving reliability and control.
- **IAM Role-Based Auth** – EC2 instances authenticate using temporary credentials, removing the need for long-lived access keys on servers.

The solution improves security, scalability, and maintainability by eliminating hardcoded credentials, protecting sensitive information through encryption, and leveraging cloud-native object storage for application assets.